

Relativity Mnemonics

Jean Zhang

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1 Introduction

One problem that I've encountered when doing relativity psets is remembering some of the complicated formulas such as how to calculate Christoffel symbols from derivatives of metric components and how to calculate the Riemann curvature tensor from the Christoffel symbols/derivatives of Christoffel symbols. Well, it is true that you can always look these formulas up, but every website/textbook/person has a different way of labeling their indices. As a result, I find memorizing the formulas to be much more efficient than looking them up. Below are two mnemonic tricks I developed to help myself remember these formulas and calculate more quickly.

2 The Christoffel Triangle

The full formula goes like this:

$$\Gamma_{\alpha\delta}^{\beta} = \frac{1}{2}g^{\beta\mu}(g_{\alpha\mu,\delta} + g_{\mu\delta,\alpha} - g_{\alpha\delta,\mu}).$$

At first it seems like a headache to remember how the indices change, but if you observe carefully you can see a very geometric pattern of how the indices swap.

The first observation that we make here is that often times, we are dealing with diagonal metrics. This means that only components like $g_{00}, g_{11}, g_{22}, g_{33}$ survives. Since $g^{\mu\nu}$ is the inverse of $g_{\mu\nu}$, we also know that only components like $g^{00}, g^{11}, g^{22}, g^{33}$ survives.

Hence, we can happily get rid of the μ on the right hand side and replace it with β , because we know that only $g^{\beta\beta}$ survives.

$$\Gamma_{\alpha\delta}^{\beta} = \frac{1}{2}g^{\beta\beta}(g_{\alpha\beta,\delta} + g_{\beta\delta,\alpha} - g_{\alpha\delta,\beta}).$$

After rewriting the equation in this form, we can observe that all the indices on the right hand side are just permutations of the indices of the Christoffel symbol. This simplifies our life a lot! Why? Normally, to calculate the Christoffel symbols, we pre-compute the derivatives of the metric. Therefore, to see if a Christoffel symbol is nonzero, we can just refer back to our list of metric derivatives and see if we can find a nonzero term.

Now we can make a second observation and see that the permutations look like walking from different vertices of a right triangle. The first term $g_{\alpha\beta,\delta}$ has a permutation pattern that looks like walking from the right angle vertex, up, and to the right. The second term $g_{\beta\delta,\alpha}$ has a permutation pattern that looks like walking from the highest vertex of the right triangle to the lower right vertex, then walking left to the right angle vertex. The third term $g_{\alpha\delta,\beta}$'s path looks like the second term's path, but in reverse. Hence, just by remembering the order of how to walk the "right triangle" of the Christoffel symbol, we can quickly remember the permutation of the indices of the metric derivatives on the right hand side.

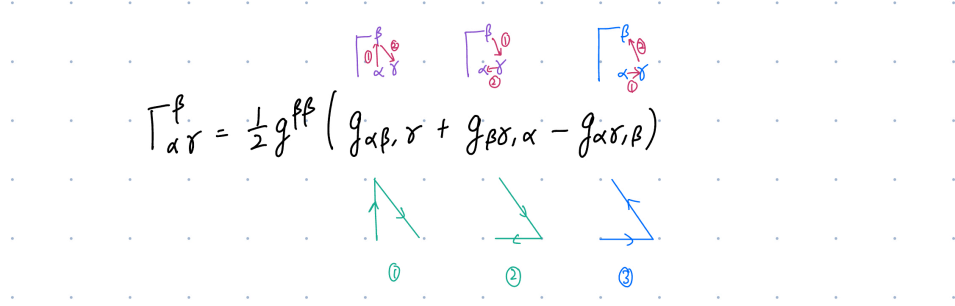


Figure 1: The Christoffel Triangle

This also allows you to see in about 10 seconds whether a particular Christoffel symbol is zero. For example, if we are in the weak field metric and we want to see if Γ_{xt}^x is zero or not, we can immediately use our triangle method to know which metric derivatives to check (term one corresponds to $g_{xx,t}$, term two corresponds to $g_{xt,x}$ and term three corresponds to $g_{xt,x}$). We can immediately see that term two cancels with term 3, and that term 1 (which we've already calculated) is 0. So you can see within 10 seconds whether a component is zero.

3 The Riemann Curvature Song

The Riemann Curvature tensor is computed from the following:

$$R_{\beta\delta\gamma}^{\alpha} = \Gamma_{\beta\gamma,\delta}^{\alpha} - \Gamma_{\beta\delta,\gamma}^{\alpha} + \Gamma_{\mu\delta}^{\alpha} \Gamma_{\beta\gamma}^{\mu} - \Gamma_{\mu\gamma}^{\alpha} \Gamma_{\beta\delta}^{\mu}$$

This also seems like a headache to remember, but again we can spot some patterns within the index permutations.

I like to label the indices from left to right (α to γ) as 0, 1, 2, 3. Now let's observe the right hand side. We see that the upper index is always α , so we don't have to worry about it. Let's now focus on the lower indices.

In the first term, $\Gamma_{\beta\gamma,\delta}^{\alpha}$, the 2nd and 3rd indices got swapped (the ordering on the LHS is $\delta\gamma$ but now on the RHS it is $\gamma\delta$). In the second term, the index ordering is the same as the left hand side. In the third term and fourth term there exists an extra contracted index μ . We can just remember that μ lives in the lower left of the first Christoffel and contracts with the upper index of the second Christoffel. There's not much we can say about that. But for the lower indices, in the third RHS term the 1st and 2nd term (β and δ) gets swapped, and in the fourth RHS term the lower indices undergo a cyclic shift to the right (from $\beta\delta\gamma$ to $\gamma\beta\delta$).

So, what changed in each term? 2& 3 (first RHS term), nothing (second RHS term), 1&2 (third RHS term), cyclic shift to the right (fourth RHS term).

I thought it would be helpful to represent these numbers as music notes (I'll use Solfège notation because it is more natural for me to remember). 1 is do, 2 is re, 3 is mi, etc. In music, la or 6 is called the reference pitch, so I thought I can use 6 to represent the second RHS term, because the second RHS term has the "standard" index pattern (there is no permutation involved). Now I just need one more number to represent the cyclic shift to the right. While 7 seems to be the natural choice (it is one unit to the right of 6), I choose 5 or so because there happens to be a song that I know that has the melodic pattern: 23-61-2-5 (re mi la do re so).

Note: This song is called 《陌上桑》 (its lyrics came from 汉乐府 (Yuefu Poetry of the Han Dynasty)).

(here's a video I can find:

<https://youtu.be/tQDevcq78S4?si=NgN1MA5xbJytceae>

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The opening lyric is “日出东南隅，照我秦氏楼”，and it has the corresponding melody [2-3-6-1-2-5]-3-1-2-3-7-5-6. I was very surprised to find a connection between the Riemann curvature tensor and the opening melody of this song.

This is a much more particular/personal mnemonic trick (I acknowledge it might be difficult for non-Chinese speakers to appreciate hhhh), but I am sure in the plethora of songs that exist on this planet people can find a melody with the corresponding pattern.